

DEM 601 Introduction to Demographic Analysis

Lecture 3 handout

TOPIC
Population Change

SESSION
8 October 2026

NAME

The shape of the lecture, not a substitute for it. Every heading is a slide, in the order it is shown, with the formulas it carries and the terms it defines.

Opening

WHERE WE ARE Setting the numbers in motion

Components of Change

COMPONENTS Closing the equation on a real year

COMPONENTS The same equation, as rates

Growth rate Everything that moved the number

Rate of natural increase Births minus deaths

Net migration rate Arrivals minus departures

MEASURES Absolute change

$$\Delta P = P_t - P_0$$

THE WORLD People added each year

Sketch the shape, and label the axes

THE WORLD Twelve thousand years, one corner

Sketch the shape, and label the axes

MEASURES Percentage change

$$\% \Delta P = \frac{P_t - P_0}{P_0} \times 100$$

MEASURES The same percentage, twice

YOUR TURN Which one grew faster?

Growth Rates

GROWTH Two counts, and a guess about the middle

Arithmetic a constant number

Geometric a constant ratio, once a year

Exponential a constant ratio, always

GROWTH Arithmetic growth

$$P_n = P_0 + (n \times (r \times P_0))$$

$$r = \frac{P_n - P_0}{n} \div P_0 \times 100$$

GROWTH Geometric growth

$$P_n = P_0 \times (1 + r)^n$$

$$r = \sqrt[n]{\frac{P_n}{P_0}} - 1$$

GROWTH Exponential growth

$$P_n = P_0 \times e^{r \times n}$$

$$r = \frac{\ln(P_n/P_0)}{n}$$

$$r(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta N(t)}{N(t) \Delta t} = \frac{\frac{dN(t)}{dt}}{N(t)} = \frac{d \ln[N(t)]}{dt}$$

GROWTH Nothing grows exponentially for ever

Sketch the shape, and label the axes

GROWTH Running the formula forwards

$$P_n = P_0 \times e^{r \times n}$$

GROWTH Where the world actually is

Sketch the shape, and label the axes

GROWTH The world and Türkiye

Sketch the shape, and label the axes

GROWTH One country, eighty-one answers

Sketch the shape, and label the axes

GROWTH Every country, one year

Sketch the shape, and label the axes

GROWTH The same year, without migration

Sketch the shape, and label the axes

YOUR TURN Find your own population

THE ANSWER Türkiye, on the two maps

Doubling Time

DOUBLING How long until twice as many?

$$2P_0 = P_0 \times e^{r \times n}$$

$$\ln(2) = r \times n$$

$$n = \frac{\ln(2)}{r}$$

DOUBLING A statement about today

Sketch the shape, and label the axes

DOUBLING What the UN actually publishes

Sketch the shape, and label the axes

YOUR TURN Double Türkiye

THE ANSWER Türkiye, doubled

Replacement and Momentum

REPLACEMENT Standing still is work

Population Turnover = Losses + Gross Growth

Sketch the shape, and label the axes

REPLACEMENT A little more than two

MOMENTUM Below replacement, still growing

Sketch the shape, and label the axes

MOMENTUM Türkiye, step by step

Fertility fell children per woman, below replacement, and it has been falling for decades _____

Births still beat deaths Natural increase a year, from many potential mothers and comparatively few old people _____

The population grows Median age , and of the country is not yet 15 _____

YOUR TURN Thirty years from now

MOMENTUM How much is already decided

Sketch the shape, and label the axes

CLOSING What you can do now